

VISHAL SEELAM

+1 (682) 374-6021 | vishalseelam21@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI professional with experience deploying AI-powered systems in healthcare and fintech. Co-founded an oncology-focused AI startup, leading end-to-end AI and security-focused development with an emphasis on user impact and business value. AWS Certified - Machine Learning Engineer skilled in building production-grade AI systems, model evaluations, and secure AI infrastructure.

PROFESSIONAL EXPERIENCE

Vaccinity Inc.

Co-Founder & AI Engineer

Sep 2024 - Present

Fort Worth, TX, USA

- Built an AI-driven clinical trial matching platform to improve patient access to oncology research; *partnered with 4+ research centers, supported 5,000+ patients - securing \$230k Health Wildcatters backing and finalist status for a \$300k SBIR grant.*
- Designed a unified AI backend that powered both a web portal and a multilingual voice assistant(WebRTC), enabling patients to complete intake via questionnaire or phone and receive real-time trial matches.
- Engineered end-to-end eligibility matching pipelines, including FHIR-based EHR ingestion, knowledge graph construction, and LLM agent workflows that retrieved and ranked trials from integrated databases (e.g., ClinicalTrials.gov) to surface personalized matches.
- Developed latency-aware orchestration workflows that optimized model inference and API throughput by parallelizing trial processing across dynamic worker pools and managing rate limits at scale.
- Led model evaluation, compliance, and monitoring efforts, deploying patient-facing LLMs with hallucination detection, prompt logging, multilingual QA (>500 calls at >85% accuracy), and HIPAA/21 CFR Part 11-compliant infrastructure.

VIInc.

ML Engineer

May 2023 - May 2024

Markham, ON, Canada

- Developed ML workflows to categorize spending data from NFC taps and scanned receipts, powering a paperless receipt system.
- Built an OCR and NLP pipeline using AWS Textract and spaCy, cutting data processing time by 40%.
- Fine-tuned BERT transformers for transaction categorization and entity recognition, improving classification accuracy by 28%.
- Implemented scalable ML workflows using TensorFlow on AWS SageMaker and orchestrated them with Apache Airflow.
- Delivered real-time analytics and personalized insights, boosting daily active usage (DAU) by 140%.

Texas Christian University

AI Research Assistant

Jan 2024 - May 2024

Fort Worth, TX, USA

- Conducted fine-tuning of LLaMA 3 (8B) using LoRA adapters on a curated corpus of 250+ de-identified clinical intake and discharge summaries, optimizing for domain alignment in health literacy audit tasks under low-resource constraints.
- Built a custom Retrieval-Augmented Generation (RAG) pipeline: Sentence-BERT embeddings, FAISS indexing (flat & HNSW), and hybrid dense + keyword retrieval for high-recall evidence search.

SKILLS

- **Core Areas:** Machine Learning, Deep Learning, Transformers, Natural Language Processing (NLP), Large Language Models (LLMs), AI Agents, Prompt Engineering, Finetuning, Retrieval Augmented Generation (RAG), Model Evaluation, MLOps, Recommender Systems, Cloud, Security, Predictive Analytics
- **Languages/Frameworks:** Python, Javascript, Typescript, Java, SQL, FastAPI, Flask, React.js, Node.js
- **AI/ML:** LangChain, LangGraph, Scikit-learn, Pytorch, Tensorflow, Hugging Face, XGBoost, MLflow, DVC, CrewAI, LlamaIndex
- **Cloud & DevOps:** AWS, Azure, GCP, Git, Github, Docker, Kubernetes, Redis, Hadoop, Airflow, Kafka, Prometheus

PROJECTS

EvalX: Advanced AI Evaluation Framework - [Link to project](#)

- A comprehensive evaluation framework supporting over 18+ metrics, including traditional NLP scores and LLM-based assessments across text, vision, and audio modalities.
- Developed a meta-evaluation system to enhance metric reliability and validity, providing industry-leading insights into AI model performance and ensuring robust, bias-free evaluations.
- Automated evaluation pipelines using agent orchestration, reducing setup time by 50% and enabling real-time testing; 1000+ users.

EDUCATION

Texas Christian University | *Bachelor's, Data Science*